

DATA 505: Homework 2

Weeks 2 and 3 (classes and types; data frames; descriptive statistics and summarizing data)

2025-09-16

Context

In this homework, you will reinforce concepts from weeks 2 and 3. Using what you learned in week 2, you will practice inspecting, subsetting, and cleaning data, as well as understanding how R handles different types of data. Then, using what you learned in week 3, you will practice producing descriptive summaries of the variables in a dataset.

References

- [Lecture 2](#)
- [Lecture 3](#)
- Chapter 5 of: Grolemund, G. (2014). Hands-On Programming with R. O'Reilly Media. <https://rstudio-education.github.io/hopr/>
- Chapter 5 of: Navarro, D. J. (2019). Learning statistics with R: A tutorial for psychology students and other beginners (Version 0.6.1). University of Adelaide. Retrieved from <https://learningstatisticswithr.com>. [Link to pdf](#)

Coding Tasks

Create an R script and use it to answer the following questions.

Part 1: Data frames, types, classes, coercion

1. Reading and Inspecting Data

In this lab we will explore a dataset containing size measurements for 3 species of penguins from the Palmer Archipelago in Antarctica.

- a. Use the command like below to read the penguins data into R as a data frame called `penguins` (see the link above).

```
penguins <- read.csv(  
  "https://raw.githubusercontent.com/allisonhorst/palmerpenguins/refs/heads/main/inst/extdata/ptarmigan.csv"  
)
```

- b. Use `class()`, `typeof()`, and `str()` to inspect the `penguins` object. What do you notice about the types and classes of the columns?
- c. Use `names()` to list all column names in the data frame.

2. Subsetting and Accessing Data

- a. Use the `$` operator to extract the `body_mass_g` column and store it as a new object called `body_mass`. What is the type of this object?
- b. Use the `[["body_mass_g"]]` syntax to extract the same column. Is the result the same as above?
- c. Use `length()` to determine how many body mass values are in the data set.
- d. Extract the body mass of the first penguin using vector subsetting.
- e. Use logical subsetting to extract the body mass values for all penguins on the "Dream" island.

3. Working with Types and Coercion

- a. Use `typeof()` to check the type of the `body_mass` vector. If it is not numeric, use `as.numeric()` to coerce it and assign the result back to `body_mass`.
- b. Update the `penguins` data frame so that the `body_mass_g` column is numeric (if it is not already).
- c. Use `typeof()` and `class()` to check the type and class of the `species` column. If it is not a factor, convert it using `as.factor()`.
- d. Use `levels()` to list the possible values of `species`.

4. Data Frame Operations

- a. Create a new data frame containing only penguins with a body mass of at least 4000 grams.
- b. Create a new data frame containing only the `species` and `island` columns for penguins with a non-missing body mass.
- c. How many penguins have missing (NA) values for body mass? Use `is.na()` and `sum()`.

5. Challenge: Creating a Categorical Variable

- a. Create a new factor column in the `penguins` data frame called `mass_group` with three levels:
 - “light” for body mass less than 3500 grams
 - “medium” for body mass between 3500 and 4500 grams (inclusive)
 - “heavy” for body mass greater than 4500 grams
- b. Display the `species`, `body_mass_g`, and `mass_group` columns for all penguins.

Part 2: Descriptive statistics and summarizing data

1. Exploring distributions

- a. For the `body_mass_g` variable:
 - Calculate the mean, median, and standard deviation.
 - What are the minimum and maximum values?
 - What are the 25th and 75th percentiles (first and third quartiles)? What is the interquartile range (IQR)?
 - Create a histogram of `body_mass_g`.
 - Write a brief description of the distribution: Is it symmetric, skewed, or does it have outliers?

2. Summarizing categorical variables

- a. Use `table()` to create a frequency table for the `species` variable.
- b. Use `prop.table()` to calculate the proportion of each species.
- c. Describe the distribution of species in the dataset.

3. Challenge: Comparing groups

- a. Calculate the mean and median `body_mass_g` for each species (hint: use `tapply()` or `aggregate()`).
- b. Create side-by-side boxplots of `body_mass_g` by species.
- c. Write a short interpretation: Which species tends to be heavier? Are there differences in spread or outliers?